Harshavardhan Kuthadi

(602) 741-4354 | hkuthadi@asu.edu | linkedin.com/in/hkuthadi | github.com/harsha0602 | Portfolio

PROFESSIONAL SUMMARY

Software engineer with 3+ years of experience building web and mobile apps, backend services, and data pipelines. Background in Python, TypeScript, C#, SQL, cloud infrastructure, and machine learning, with an M.S. in Software Engineering from Arizona State University.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#, Java, Scala, C++, HTML, CSS

Frontend & backend: React, React Native, Next.js, FastAPI, Node.js, .NET

Cloud & data: AWS, Azure AKS, Docker, PostgreSQL, pgvector, MSSQL, MongoDB

AI & testing: OpenAI Agents SDK, OpenClaw, PyTorch, scikit-learn, pytest, Playwright, Vitest, GitHub Actions

WORK EXPERIENCE

Pretorin May 2026 to Present
Product Engineer *Tempe, AZ*

- Own development of Compliance Claw and the in-app assistant that helps people navigate Pretorin and get support.
- Work on MCP hosting and plugins so people can use Pretorin's tools from OpenAI, Anthropic, and other AI clients.
- Build Python and FastAPI services that check permissions, look up relevant evidence, and keep responses linked to their sources. Add tests for denied access and prompt injection.
- Improve the agent development stack and CLI releases. Added signed packages, checksums, and software bills of materials; reduced macOS installation time from seven minutes to three.

TappedIn Nov 2025 to May 2026
Technical Co-founder *Remote, US*

- Built artist discovery, user portfolios, trading, and account flows with React Native and Expo.
- Keep balance changes and trades in a single database transaction, with checks that prevent a repeated request from charging twice.
- Batch Spotify artist updates through a queue and retry failed requests. Added tests for the app, APIs, database permissions, and trading logic.

Handshake AI Solutions, LLC Sep 2025 to Oct 2025
AI Trainer - Computer Science Expert *Remote, US*

- Wrote more than 50 computer science prompts and reviewed more than 100 model answers for technical correctness, edge cases, and code errors.

Amazon Web Services (AWS) May 2025 to Aug 2025
Software Development Engineer Intern *Seattle, WA*

- Built AWS Glue jobs in Scala and PySpark to turn JSON and DynamoDB exports into partitioned Parquet on S3. Reduced data preparation time by about 95%.
- Moved prototypes into Scala, fixed crawler timing and job-bookmark issues, and adjusted compute settings to cut runtime by 50%.
- Set up infrastructure with AWS CDK and CloudFormation. Added CloudWatch alerts and dashboards for more than 100 active streams.
- Compared storage and processing options and wrote design documents covering costs, schemas, and partitioning.

o9 Solutions, Inc. Jul 2022 to Jul 2024
Software Development Engineer *Bengaluru, India*

- Built configuration screens for visualization widgets so consultants could set up dashboards without writing custom code.
- Developed Git-style versioning and conflict resolution for tenant configurations used by more than 20 teams.
- Refactored Jasmine tests and the code they covered, removing more than 10,000 lines of duplicate code. Fixed frontend and backend bugs across releases.

OpenNets May 2021 to May 2022
Software Developer *Bengaluru, India*

- Used Mininet, Node.js, and MongoDB to model and validate more than 10 network topologies.
- Built an AngularJS interface for editing settings and viewing network layouts.
- Added automated configuration checks to catch errors before running a simulation.

SELECTED PROJECTS

FluidFill | Code | Demo Nov 2025

Python | FastAPI | Next.js | TypeScript | Postgres | DOCX | Gemini API | Express

Fill out a document once, even when the same field appears in several places.

- Built a Next.js form editor and Python services that detect placeholders and generate document previews.
- Used Gemini to identify fields, with a parser fallback when the model is unavailable. Kept repeated fields in sync and saved form answers in PostgreSQL.

Realtime Study Rooms | Code | Demo Oct 2025

React | Node.js | WebSockets | WebRTC | TypeScript | ws

Study rooms with chat, online status, and video calls.

- Built a Node.js WebSocket server for rooms, messages, and the connection setup needed for WebRTC video calls.
- Wrote React hooks for connection management and media streams, plus an 11-step tutorial and starter code for other students.

Clozyt | Code | Demo Sep 2025

TypeScript | PyTorch | Tailwind CSS | Qdrant

A shopping app that learns from the clothes you like and skip.

- Built the shopping interface with Next.js and Tailwind CSS, including a cart and checkout flow.
- Used FashionCLIP and Qdrant to find similar products, mixed in new suggestions, and built APIs that record feedback and return the next batch of items.

FluxRate | Code | Demo Aug 2025

Java | Maven | Python | pandas | matplotlib | CPS Design | UML

A parking simulation with gate controls and prices that change with demand.

- Implemented gate states, timing checks, emergency handling, and payment checks for exiting vehicles.
- Ran a 24-hour scenario with 50 spaces and used Python plots to check occupancy, billing, gate behavior, and price changes.

Safe Python Execution Service | Code Mar 2025

Python | Flask | nsjail | Docker | GCP Cloud Run

Run submitted Python code with limits on what it can do.

- Disabled network access and set limits on execution time and system resources.
- Packaged the service for Google Cloud Run so it can stop when no requests are running.

Market Risk Forecasting Dashboard | Code Apr 2025

Python | TensorFlow | AWS SageMaker | SQL

Experiments in forecasting market volatility from historical price data.

- Processed historical price and trading-volume data with Python and SQL, including rolling volatility, moving averages, and correlations.
- Trained TensorFlow models and deployed an inference workflow on AWS SageMaker.

EDUCATION

Arizona State University, Tempe, AZ Aug 2024 to May 2026

Master's in Software Engineering | GPA: 3.67/4

National Institute of Technology Karnataka, Surathkal, India Jul 2018 to May 2022

Bachelor of Technology in Electronics and Communications Engineering

ACHIEVEMENTS

- Visual AI Hackathon Winner: Won with the Food Sage team at ASU. We built an app that scans food labels, extracts ingredients, and flags potential ingredient risks.
- 2x Performer of the Quarter: Awarded Performer of the Quarter twice at o9 Solutions.